RIPHAH INTERNATIONAL COLLEGE
ADP (ITM/CS/A&F/BM/Commerce)

Course Title:	**Finance for Computer Science**	**Semester:**	**IV**
Course code:	**FIN-201**	**Credit Hours:**	**3**
Prerequisites	Financial accounting-I	**Lecture type**	**Classroom Lectures/Presentation/ case studies**

Instructor Name: **Email:**
Consulting Hours: **Contact info**

I. Course Description:

Financial markets form the source of a vast number of challenging computational problems. These are not only intellectually challenging from the point of view of computational modelling, but the financial sector is also an employer of a significant fraction of students of Computer Science. This course covers the use of computational processing in Finance such as the use of computation in the analysis of finance algorithmic instruments.

II. Course Objectives

This course will help students to:

- To understand why Return & risk are the critical components of all investment decisions
- To identify the importance & mechanism of money market and capital market securities for investment decisions
- To distinguish between the direct & indirect investment opportunities
- To understand how the securities are traded.
- To recognize the modern portfolio theory
- To understand the valuation of securities for investment analysis purpose
- To understand investment practices in global environment
- To understand the importance of ethical guidelines for sustainable investments

III. Course Learning Outcomes:

After studying this course, students should be able to:

- Identify the primary goal of finance to maximize shareholder wealth and not profits and its relationship to decision-making within the firm.
- Identify key components of the financial market system and the role of different market players, costs, vehicles, efficient markets, interest rate determinant, and risk-return tradeoffs.
- Evaluate a company's financial performance, using a comprehensive set of financial tools.
- Analyze the use of time value of money in present value and future value models for investment/project decision-making purposes.
- Define and measure the expected rate of return and risk for an individual investment, to explain the relationship between risk and return, and how diversification affects risk and

IV-Course Grading Policy:

The instructor is responsible for grading all student performance through examinations, class participation in discussions, individual and/or team presentations, short or major papers requiring research

or analysis, and other appropriate means. Individual discussions of the reading material may be held at the instructor's discretion. If any projects or assignments are identical or partially identical, a zero will be awarded. The repetition of such instances may result in an "F" grade in the course

Grade Distribution:

Evaluation Type	Percentage (%)	Activities
Quizzes	10	Minimum 4
Assignments	10	Minimum 4
Presentation	10	Minimum 1
Mid Term	30	
Final Term	40	
Total Points	**100**	

ABSOLUTE GRADING

V-General Classroom Norms (Attendance required: Minimum 75%)

Class attendance is mandatory. You may miss up to **25%** (8 out of 32 sessions) of class sessions, but save it for emergencies only. In case you exceed this level, you will be withdrawn from the course. As a courtesy to the instructor and other students, be prepared to arrive at class and be in your seat on time. In addition, please note that each class lasts for **90 minutes** (1.5 Hours).

Also, keep in mind some general rules in the class as given below:

- Cell phones should be powered off.
- Eatables are not allowed in the class.
- The teacher will not tolerate any disruptive behavior in the class.

The University Dress Code has to be observed; no warnings will be given, and violators will be asked politely to leave the class and consequently will be marked absent.

VI-Statement on Faculty and Student Code of Ethics

The Code of Ethics establishes the standards for proper conduct in the classroom for both faculty and students. Although this is a foundational concept, it emphasizes the key principle we will uphold as a team. The faculty member will do their best to create an optimal teaching and learning environment for each student individually and for the class as a whole. Students will receive a fair evaluation for all their work and their contributions to achieving the course objectives. On the other hand, students will be responsible for their own actions, providing proper documentation when they use the work of others as references. They will also respect the rights of others to contribute to the teaching/learning environment according to their capabilities and motivations.

VII-Course Content & Schedule:

Week	Contents	Activities
1	**An Introduction to the Foundations of Financial Management:** • Five Principles That Form the Foundations of Finance • The Role of Finance in Business • The Legal Forms of Business Organization	**Theory**
2-3	**Understanding Financial Statements and Cash Flows:** • The Income Statement • Measuring Cash Flows	**Assignment 1 (Week 2)** **Quiz 1 (Week 3)** **Numerical**
4	**Evaluating a Firm's Financial Performance** • The Purpose of Financial Analysis • Measuring Key Financial Relationships • The Limitations of Financial Ratio Analysis	**Theory**
5-6	**The Time Value of Money** • Compound Interest, Future, and Present Value • Annuities • Making Interest Rates Comparable • The Present Value of an Uneven Stream and Perpetuities	**Assignment 2 (Week 5)** **Quiz 2 (Week 6)** **Theory + Numerical**
7-8	**The Meaning and Measurement of Risk and Return** • Expected Return Defined and Measured • Risk Defined and Measured • Rates of Return: The Investor's Experience • Risk and Diversification • The Investor's Required Rate of Return	**Theory + Numerical**
MID TERM		
10	**The Valuation and Characteristics of Bonds** • Types of Bonds • Terminology and Characteristics of Bonds • Defining Value • What Determines Value? • Valuation: The Basic Process • Valuing Bonds • Bond Yields • Bond Valuation: Three Important Relationships	**Assignment 3** **Theory + Numerical**
11	**The Valuation and Characteristics of Stock** • Preferred Stock • Valuing Preferred Stock • Common Stock • Valuing Common Stock • The Expected Rate of Return of Stockholders	**Theory + Numerical**

12-13	**Capital-Budgeting Techniques and Practice** • Finding Profitable Projects • Capital-Budgeting Decision Criteria • Capital Rationing • Ranking Mutually Exclusive Projects	**Quiz 3 (Week 12)** **Theory + Numerical**
14-15	**Short-Term Financial Planning** • Financial Forecasting • Limitations of the Percent of Sales Forecasting Method • Constructing and Using a Cash Budget	**Assignment 4 (Week 14)** **Quiz 4 (Week 15) Theory**
16-17	**Working-Capital Management** • Managing Current Assets and Liabilities • Determining the Appropriate Level of Working Capital • The Cash Conversion Cycle • Estimating the Cost of Short-Term Credit Using the Approximate Cost-of-Credit Formula • Sources of Short-Term Credit	**Theory + Numerical**
FINAL TERM		

VIII. Course Material:

a.Recommended Textbooks:

1. Principles of managerial Finance by ***Lawrence j.Gitmen*** (12 th ed.)
2. Financial Management theory and practice by . ***Michael Ehrhardt***
3. Corporate Financial management ***by Glen Arnold and Deborah lewis***

b. Supplementar

- PPT and Lecture notes and other content can be provided
- Use MS-excel
- At https://drive.google.com/drive/folders/14S1kaKYPv4sR6k8wc_rq0kv3mmMaEppU? usp=drive_link

c. ICT R&F Proposal plan

d. Smeda (Small And Medium Enterprise Development Authority) https://smeda.org/